API Endpoint Plan

Allowed application roles: Organiser and Participant. Public and Authenticated describe access levels, not user roles.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account. Password is hashed before storage.	Public	FirstName, LastName, Email, Password	201 Created. Returns UserID, name, email, role and CreatedAt. 400 Bad Request. 409 Conflict if email is already registered.
POST	/api/auth/login	Authenticates a user and returns an authentication token.	Public	Email, Password	200 OK. Returns access token, UserID, name, email and role. 400 Bad Request. 401 Unauthorized for invalid credentials.
GET	/api/users/me	Retrieves the profile of the currently authenticated user.	Authenticated	None	200 OK. Returns UserID, FirstName, LastName, Email, Role, CreatedAt. 401 Unauthorized.
PUT	/api/users/me	Updates the currently authenticated user's profile information.	Authenticated	FirstName, LastName, Email	200 OK. Returns updated profile. 400 Bad Request. 401 Unauthorized. 409 Conflict if email is already in use.
PUT	/api/users/me/password	Changes the authenticated user's password.	Authenticated	currentPassword, newPassword	200 OK. Returns password update confirmation. 400 Bad Request. 401 Unauthorized.
GET	/api/users/{userId}	Retrieves a specific user's profile. Useful for viewing organiser information.	Organiser	None	200 OK. Returns UserID, FirstName, LastName, Email, Role and CreatedAt. 401 Unauthorized. 403 Forbidden. 404 Not Found.
GET	/api/users	Retrieves a list of registered users.	Organiser	None	200 OK. Returns a list of users. 401 Unauthorized. 403 Forbidden.
GET	/api/events	Retrieves all available events.	Public	None	200 OK. Returns event name, description, date, venue, organiser and creation date.
GET	/api/events/{eventId}	Retrieves detailed information about one event.	Public	None	200 OK. Returns event details, venue, organiser and categories. 404 Not Found.
POST	/api/events	Creates a new event. The authenticated user becomes the organiser.	Organiser	VenueID, Name, Description, EventDate	201 Created. Returns EventID and event details. 400 Bad Request. 401 Unauthorized. 403 Forbidden. 404 Not Found if VenueID does not exist.
PUT	/api/events/{eventId}	Updates an existing event.	Organiser	VenueID, Name, Description, EventDate	200 OK. Returns updated event. 400 Bad Request. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found.
DELETE	/api/events/{eventId}	Deletes an event.	Organiser	None	204 No Content. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found.
GET	/api/events/{eventId}/enrolments	Retrieves all participants enrolled in an event.	Organiser	None	200 OK. Returns enrolment records with participant and category information. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found.
GET	/api/events/{eventId}/categories	Retrieves all categories belonging to an event.	Public	None	200 OK. Returns categories associated with the event. 404 Not Found.
GET	/api/categories/{categoryId}	Retrieves details for a specific category.	Public	None	200 OK. Returns CategoryID, EventID, Name, DistanceKm, MaxParticipants and EntryFee. 404 Not Found.
POST	/api/events/{eventId}/categories	Creates a category for an event.	Organiser	Name, DistanceKm, MaxParticipants, EntryFee	201 Created. Returns CategoryID and category details. 400 Bad Request. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/categories/{categoryId}	Updates an existing event category.	Organiser	Name, DistanceKm, MaxParticipants, EntryFee	200 OK. Returns updated category. 400 Bad Request. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found.
DELETE	/api/categories/{categoryId}	Removes a category from an event.	Organiser	None	204 No Content. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found. 409 Conflict if category has existing enrolments.
POST	/api/events/{eventId}/enrolments	Enrols the authenticated participant into a category for an event.	Participant	CategoryID	201 Created. Returns EnrolmentID, UserID, CategoryID, EnrolmentDate and Status. 400 Bad Request. 401 Unauthorized. 404 Not Found. 409 Conflict if already enrolled or category is full.
GET	/api/enrolments	Retrieves the authenticated user's enrolments.	Participant	None	200 OK. Returns a list of the user's enrolments. 401 Unauthorized.
GET	/api/enrolments/{enrolmentId}	Retrieves details about a specific enrolment.	Participant, Organiser	None	200 OK. Returns enrolment details. 401 Unauthorized. 403 Forbidden if participant does not own the enrolment. 404 Not Found.
PUT	/api/enrolments/{enrolmentId}	Updates the status of an enrolment.	Organiser	Status	200 OK. Returns updated enrolment. 400 Bad Request. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found.
DELETE	/api/enrolments/{enrolmentId}	Cancels an enrolment.	Participant, Organiser	None	204 No Content. 401 Unauthorized. 403 Forbidden if participant does not own the enrolment. 404 Not Found.
POST	/api/enrolments/{enrolmentId}/results	Records the result for a participant after an event.	Organiser	FinishTime, Position	201 Created. Returns ResultID, EnrolmentID, FinishTime, Position and RecordedAt. 400 Bad Request. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found. 409 Conflict if result already exists.
GET	/api/results/{resultId}	Retrieves a specific result.	Participant (own result), Organiser	None	200 OK. Returns result information and associated enrolment. 401 Unauthorized. 403 Forbidden if participant does not own this result. 404 Not Found.
PUT	/api/results/{resultId}	Updates a participant's result.	Organiser	FinishTime, Position	200 OK. Returns updated result. 400 Bad Request. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found.
DELETE	/api/results/{resultId}	Removes a recorded result.	Organiser	None	204 No Content. 401 Unauthorized. 403 Forbidden. 404 Not Found.
GET	/api/enrolments/{enrolmentId}/results	Retrieves the result associated with a specific enrolment.	Participant, Organiser	None	200 OK. Returns ResultID, FinishTime, Position and RecordedAt. 401 Unauthorized. 403 Forbidden if participant does not own the enrolment. 404 Not Found.
GET	/api/events/{eventId}/results	Retrieves all results for an event.	Organiser	None	200 OK. Returns a ranked list of participants and their results. 401 Unauthorized. 403 Forbidden unless caller is the event's organiser. 404 Not Found.
GET	/api/users/me/results	Retrieves all results belonging to the authenticated participant.	Participant	None	200 OK. Returns the participant's results across events. 401 Unauthorized.
GET	/api/venues	Retrieves all venues.	Public	None	200 OK. Returns venue details and location information.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/venues/{venueId}	Retrieves a specific venue.	Public	None	200 OK. Returns venue name, address, city, province, latitude and longitude. 404 Not Found.
POST	/api/venues	Creates a new venue.	Organiser	Name, AddressLine, City, Province, Latitude, Longitude	201 Created. Returns VenueID and venue details. 400 Bad Request. 401 Unauthorized. 403 Forbidden.
PUT	/api/venues/{venueId}	Updates venue information.	Organiser	Venue fields to update	200 OK. Returns updated venue. 400 Bad Request. 401 Unauthorized. 403 Forbidden. 404 Not Found.
DELETE	/api/venues/{venueId}	Removes a venue.	Organiser	None	204 No Content. 401 Unauthorized. 403 Forbidden. 404 Not Found. 409 Conflict if venue has associated events.